

YEAR 11
PRELIMINARY EXAMINATION 2004

MATHEMATICS

Time Allowed – 85 minutes
(Plus 5 minutes Reading Time)

All questions may be attempted

All questions are of equal value

Department of Education approved calculators are permitted

In every question, show all necessary working

Marks may not be awarded for careless or badly arranged work

No grid paper is to be used unless provided with the examination paper

The answers to all questions are to be returned in separate bundles clearly labeled Question 1, Question 2, etc. Each question must show your Candidate's Number.

Year 11 Preliminary Exam 2004 Maths

Question 1

Marks

- a) Differentiate with respect to x , then simplify
- i. $f(x) = 3^{5x}$ 1
- ii. $f(x) = \frac{e^x}{\sin 2x}$ 2
- b) The equation of a parabola is $(x-3)^2 = -12(y-1)$. Find the:
- i. Coordinates of its vertex 1
- ii. Equation of its directrix 2
- c) i. Write down the discriminant of $x^2 - 2kx + 6k$. 1
- ii. For what real values of k is $x^2 - 2kx + 6k$ always positive? 2
- d) Given that $\tan \theta = T$ and θ is in the first quadrant, express in terms of T the value of $\sin(180^\circ + \theta)$. 2
- e) Solve $\sin x - \sqrt{3} \cos x = 0$ for $0^\circ \leq x \leq 360^\circ$ 2
- f) Solve for all real x : $|x+3| = 2x-1$ 2

Question 2 (start a new page)

Marks

a) Evaluate $\log_{27} 9$

2

b) Express as a single log: $2 \log a + \frac{1}{2} \log b - \log c$

2

c) Solve for k : $9^k (3^3)^4 = 1$

2

d) If α and β are the roots of the equation $x^2 - 2x - 3 = 0$,find the value of i. $\alpha + \beta$

1

ii. $\frac{1}{\alpha} + \frac{1}{\beta}$

2

e) Find the value of $\lim_{x \rightarrow 4} \left(\frac{x^2 - x - 12}{x - 4} \right)$

2

f) i. Find the stationary points of $y = x(\ln x)$ and determine their nature.

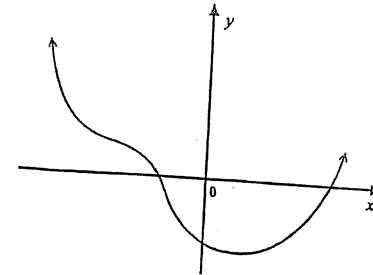
3

ii. Show that y'' of $y = x(\ln x)$ is always positive?

1

Question 3 (start a new page)

Marks

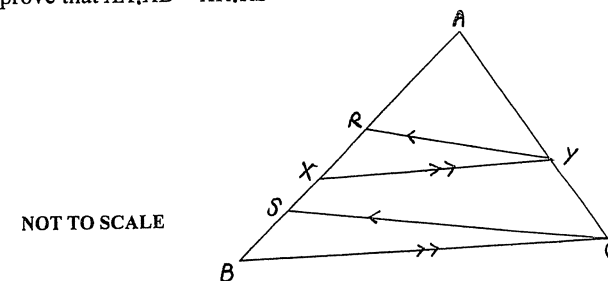
a) This is the graph of $y = f(x)$.Sketch the graph of its derivative, $y = f'(x)$ on the given set of axes.

2

(see back page)

b) Given that in $\triangle ABC$, $XY \parallel BC$, and $RY \parallel SC$,

2

prove that $AX:XB = AR:RS$ 

c) i. Write the domain of the function.

1

$$f(x) = \sqrt{3+x} - \sqrt{5-x}$$

ii. Prove that this is a monotonic increasing function between its extreme points.

2

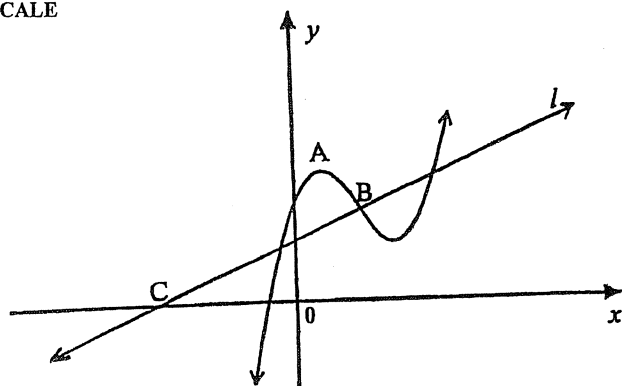
iii. Find the range of this function.

1

d)

Marks

NOT TO SCALE



The diagram shows a sketch of the curve $y = x^3 - 6x^2 + 9x + 4$.

The curve has a local maximum point at A and a point of inflexion at B .

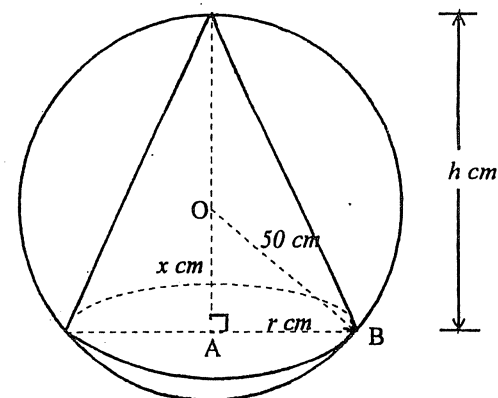
The line l is a normal to the curve at point B and meets the x -axis at point C .

- | | |
|--|---|
| i. Find the coordinates of point A . | 2 |
| ii. Show that the coordinates of point B is $(2,6)$. | 2 |
| iii. Show that the equation of the line l is $x - 3y + 16 = 0$. | 2 |
| iv. Find the coordinates of C . | 1 |

Question 4 (start a new page)

Marks

a)



NOT TO SCALE

The diagram shows a cone of base radius r cm and height h cm inscribed in a sphere of radius 50 cm. The centre of the sphere is O and $\angle OAB = 90^\circ$.

Let $OA = x$ cm.

- | | |
|--|---|
| i. Show that $r = \sqrt{2500 - x^2}$. | 1 |
| ii. Hence show that the volume, V cm ³ , of the cone is given by: | 1 |

$$V = \frac{\pi}{3}(2500 - x^2)(50 + x)$$

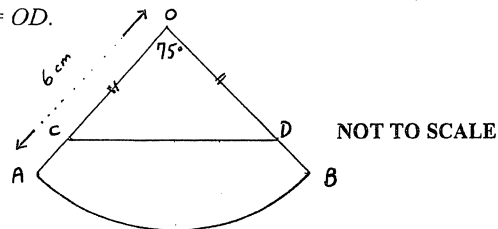
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|--|---|
| iii. Find the radius of the largest cone which can be inscribed in the sphere. | 5 |
|--|---|

b) The diagram shows a sector AOB . The length of OA is 6 cm and

Marks

$\angle AOB = 75^\circ$. The points C and D lie on OA and OB respectively

and $OC = OD$.



i. Convert 75° to radian measure, leaving the answer in terms of π . 1

ii. Find the exact area of the sector AOB . 1

iii. If the area of the triangle COD is one half the area of the 2

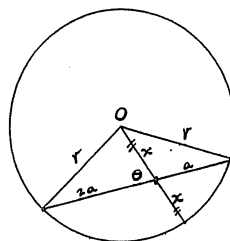
sector AOB , find the length of OC correct to two decimal places.

c) The following diagram shows a circle with centre O and radius r units.

A radius divides a chord in the ratio of 2 : 1 and is bisected by the chord

as shown in the diagram.

NOT TO SCALE



i) Show $r^2 = x^2 + 4a^2 - 4ax \cos \theta$ 1

ii) Show that the cosine of the angle θ between the chord and 3

the radius is $\frac{1}{4}\sqrt{6}$.

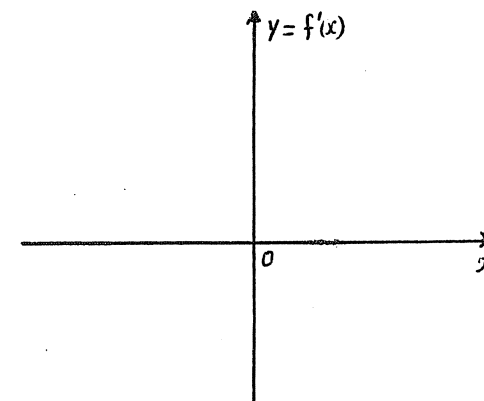
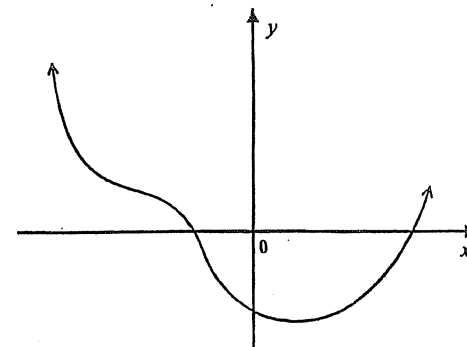
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REMEMBER TO HAND IN THIS PAGE Student Number: _____

Question 3 (a)

This is the graph of $y = f(x)$.

Sketch the graph of its derivative, $y = f'(x)$ on the given set of axes.



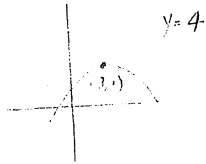
Question 1

a) i) $f(x) = (3^{5x} \cdot \ln 3) \times 5$

ii) $f'(x) = \frac{(\sin 2x) e^x - 2(\cos 2x) e^x}{(\sin 2x)^{\frac{1}{2}}}$
 $= \frac{e^x (\sin 2x - 2\cos 2x)}{(\sin 2x)^{\frac{1}{2}}}$

b)

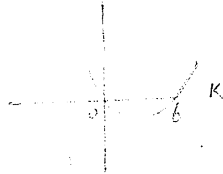
i) $(3, 1)$
 $a=3$
 ii) $y=4$



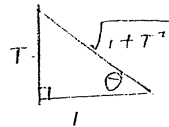
c) i) $\Delta = (-2k)^2 - 4(6k)$
 $= 4k^2 - 24k$

ii) concave up

$\Delta < 0$
 $4k^2 - 24k < 0$
 $k^2 - 6k < 0$
 $k(k-6) < 0$
 $0 < k < 6$



d) $\sin(180^\circ + \theta)$
 $= -\sin \theta$
 $= -\frac{T}{\sqrt{1+T^2}}$



e) $\sin x = \sqrt{3} \cos x$
 $\tan x = \sqrt{3}$
 $x = 60^\circ, 240^\circ$
 $\frac{1}{2}, \frac{1}{2}$

f) $|x+3| = 2x-1$

$x+3 = 2x-1$
 $4 = x$

$x+3 = -(2x-1)$
 $3x = -2$
 $x = -\frac{2}{3}$

Check $x=4$
 $|4+3| = |7| = 7$
 $2x-1 = 2(4)-1 = 7$
 $\therefore x=4$

Check $x = -\frac{2}{3}$
 $|\frac{-2}{3}+3| = |\frac{7}{3}| = \frac{7}{3}$
 $2(-\frac{2}{3})-1 = -\frac{7}{3} \neq \frac{7}{3}$
 $\therefore x \neq -\frac{2}{3}$

$\therefore x=4$

Q 2

a) $\log_{27} 9 = \frac{\log 9}{\log 27} = \frac{\log 3^2}{\log 3^3} = \frac{2 \log 3}{3 \log 3} = \frac{2}{3}$

b) $\log a^2 + \log b - \log c = \log \left(\frac{a^2 \cdot b}{c} \right)$

c) $9^k (3^3)^4 = 1$
 $3^{2k} \cdot 3^{12} = 3^0$
 $2k+12=0$
 $k = -6$

d) i) $\alpha + \beta = -\frac{b}{a} = 2$
 ii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{2}{-3}$

e) $\lim_{x \rightarrow 4} \frac{(x-4)(x+3)}{(x-4)} = 4+3 = 7$

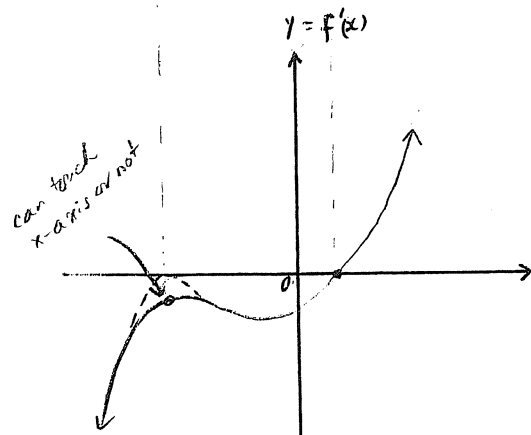
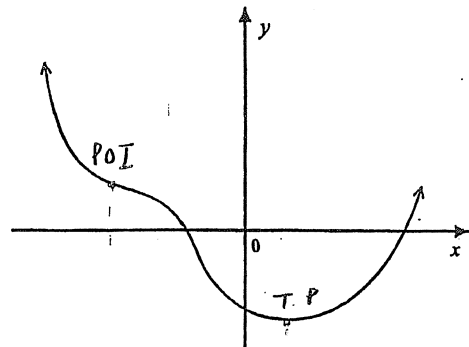
f) i) $y = x(\ln x)$
 T.P: $y' = \frac{x}{x} + \ln x = 1 + \ln x = 0$ when $-1 = \ln x$
 $\therefore \frac{1}{e} = x$
 when $x = \frac{1}{e}$, $y = \frac{1}{e} \ln \frac{1}{e} = \frac{1}{e} \ln e = -\frac{1}{e}$
 $\therefore \text{T.P} \left(\frac{1}{e}, -\frac{1}{e} \right)$

$y'' = \frac{1}{x}$
 when $x = \frac{1}{e}$, $y'' = \frac{1}{\frac{1}{e}} = e > 0$
 $\therefore \left(\frac{1}{e}, -\frac{1}{e} \right)$ is minimum

2f ii) $y'' = \frac{1}{x} > 0$ because $y = x(\ln x)$ is only defined for $x > 0$ /

Question 3

a)



2

3 b) In $\triangle ABC$

$XY \parallel BC$ (given)

$AX : XB = AY : YC$ (line parallel to one side of triangle divides the other 2 sides into same ratio)

Similarly in $\triangle ASC$

$RY \parallel SC$ (given)

$AR : RS = AY : YC$ $\frac{1}{2}$

$\therefore AX : XB = AR : RS$ $\frac{1}{2}$

c) Domain: $x \geq -3$ and $x \leq 5$
i.e. $-3 \leq x \leq 5$

ii) $f'(x) = \frac{1}{2\sqrt{3+x}} + \frac{1}{2\sqrt{5-x}} > 0$ for $-3 < x < 5$

1 m for $f'(x)$

1 m for > 0

$\therefore$ monotonically increasing between its extreme points

iii)

When $x = -3$ $f(x) = -\sqrt{8}$

$x = 5$ $f(x) = \sqrt{8}$

$\therefore$ Range: $-\sqrt{8} \leq f(x) \leq \sqrt{8}$

d) i) $y = x^3 - 6x^2 + 9x + 4$

$y' = 3x^2 - 12x + 9 = 0$

$x^2 - 4x + 3 = 0$

$(x-3)(x-1) = 0 \therefore x = 1$ or 3

$\therefore x_A = 1$ from graph A is closer to y-axis

$y_A = 1^3 - 6 \cdot 1^2 + 9 \cdot 1 + 4 = 1 - 6 + 9 + 4 = 8$

$\therefore A(1, 8)$

(USA cond.)

ii) $y'' = 6x - 12 = 0 \Rightarrow x = 2$
 $x = 2, y = 2^3 - 6 \times 2 + 18 + 4 = 8 - 12 + 22 = 6$
POI at B (2, 6)

iii) $y' = 3x^2 - 12x + 9$
At B, gradient = $y' = 3x^2 - 12x + 9 = 12 - 24 + 9 = -3$

gradient of normal at B = $\frac{1}{3}$

$\therefore$ gradient of line $l = -\frac{1}{3}$

equation of line $l: y - 6 = \frac{1}{3}(x - 2)$

$$y - 6 = \frac{x}{3} - \frac{2}{3}$$

$$3y - 18 = x - 2$$

$$x - 3y + 16 = 0$$

iv) At $y = 0, x = -16$

$\therefore C = (-16, 0)$

Q4 a) i) Using Pythagoras' Theorem

$$50^2 = r^2 + x^2$$

$$\therefore r^2 = 2500 - x^2$$

$$r = \sqrt{2500 - x^2} \quad (r \geq 0)$$

$\frac{1}{2}$ m off for not stating $r \geq 0$

ii) $V = \frac{1}{3} \pi r^2 h$ $h = 50 + x$

$$V = \frac{1}{3} \pi (2500 - x^2)(50 + x)$$

at least 1 step between

iii) $V' = \frac{\pi}{3} [(-2x)(50 + x) + (2500 - x^2)]$

$$V' = \frac{\pi}{3} [-100x - 2x^2 + 2500 - x^2]$$

$$V' = \frac{\pi}{3} [-3x^2 - 100x + 2500] = 0$$

$$3x^2 + 100x - 2500 = 0$$

$$x = \frac{-100 \pm \sqrt{10000 - 4(3)(-2500)}}{6}$$

1.5

$$x = \frac{-100 \pm \sqrt{40000}}{6}$$

$$x = \frac{50}{3} \text{ or } -50$$

But $x > 0 \therefore x = \frac{50}{3}$ only

$$r = \sqrt{2500 - \left(\frac{50}{3}\right)^2}$$

$$r = \frac{100\sqrt{2}}{3} \text{ cm or } 47.14 \text{ cm}$$

To determine max/min: Test $\frac{dv}{dx}$ as $V, \frac{dv}{dx}$ are continuous for $0 \leq x \leq 50$

x	16.6	$\frac{50}{3}$	16.7
V'	13.95	0	-6.98

$\therefore$ Relative max when $r = \frac{100\sqrt{2}}{3} \text{ cm}$.

Since there is only one turning point in the domain $0 \leq x \leq 50$ it is also the absolute max.

Alternatively assume $-50 \leq x \leq 50$

$$V' = 0 \text{ when } x = -50, \frac{50}{3}$$

Test V'' for concavity as V, V', V'' are continuous for $-50 \leq x \leq 50$

$$V'' = \frac{\pi}{3} (-6x - 100)$$

$$V''(-50) = \frac{200\pi}{3} > 0 \therefore \text{rel min at } x = -50$$

$$V(-50) = 0 \text{ cm}^3$$

$$V''\left(\frac{50}{3}\right) = -\frac{200\pi}{3} < 0 \therefore \text{rel max at } x = \frac{50}{3}$$

$$V\left(\frac{50}{3}\right) = 155140.4 \text{ cm}^3$$

Check end points $V(50) = 0 \text{ cm}^3$

$\therefore x = \frac{50}{3}$ gives absolute max

$$\text{or } r = \frac{100\sqrt{2}}{3} = 47.1 \text{ cm gives absolute max}$$

i) Area sector $AOB = \frac{1}{2} r^2 \theta = \frac{1}{2} \times 6^2 \times \frac{5\pi}{12} = \frac{15\pi}{2} \text{ cm}^2$ /
(or 23.56 cm^2)

ii) Let $OC = x \text{ cm}$

Area $\triangle COD = \frac{1}{2} x^2 \sin \frac{5\pi}{12}$

$\frac{15\pi}{2} = 2 \left[\frac{x^2}{2} \sin \frac{5\pi}{12} \right]$ /

$\frac{15\pi}{2} = x^2 \sin \frac{5\pi}{12}$

(Q4b cont'd)

$x^2 = \frac{15\pi}{2} \div \left(\sin \frac{5\pi}{12} \right)$

$x^2 = 24.3931 \dots$

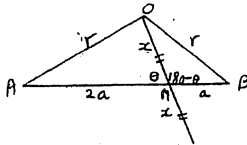
$x = 4.94 \text{ cm}$

$OC = 4.94 \text{ cm}$ (2dp)
($OC > 0$)

c) In $\triangle OAM$

$y^2 = x^2 + (2a)^2 - 2(2ax \cos \theta)$

$x^2 + 4a^2 - 4ax \cos \theta = y^2$ /



ii) Similarly in $\triangle OBM$

$y^2 = x^2 + a^2 - 2ax \cos(180^\circ - \theta) = x^2 + a^2 + 2ax \cos \theta$

$x^2 + 4a^2 - 4ax \cos \theta = x^2 + a^2 + 2ax \cos \theta$

$3a^2 = 6ax \cos \theta$

$\frac{3a}{2x} = \cos \theta \dots \text{Eq. 1}$ /

Let N be the midpoint of AB

$\therefore ON \perp AB$

$AN = BN = \frac{3a}{2}$

$(ON)^2 = r^2 - \left(\frac{3a}{2}\right)^2 = x^2 - \frac{9a^2}{4}$

$r = 2x$

$(2x)^2 - \frac{9a^2}{4} = x^2 - \frac{9a^2}{4}$

$3x^2 = 2a^2$ /

$x = \frac{\sqrt{6}}{3} a$ ($x > 0$)

Since $\cos \theta = \frac{a}{2x}$ (from Eq. 1)

$\cos \theta = \frac{a}{2 \cdot \frac{\sqrt{6}}{3} a} = \frac{\sqrt{3}}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{6}}{4}$ /



Alternatively (3u)

Products of intercepts of intersecting chords are equal

$x(3x) = 2a^2$

$3x^2 = 2a^2$

$x = \frac{\sqrt{6}}{3} a$

$\frac{r}{2} = \frac{\sqrt{6}}{3} a$